

Advanced Communication Networks

شبکه‌های ارتباطی پیشرفته

Prerequisites Review

OSI, RF, Digital Communications, IP and Decibels

مرور پیش‌نیازها



Lecture Notes

Instructor: Amin Azari

Compiled by Aidin Shekari

استاد: امین آذری github.com/AminAzari

In This Module

1	OSI Model Review (Telecom Focus)	2
2	Basic RF Concepts	3
3	Digital Communications Basics	4
4	IP Networking Fundamentals	4
5	Basic Probability	5
6	Decibel Math	6
	Summary Checklist	7

This document reviews foundational concepts assumed as prior knowledge. Use it as a refresher before diving into the course modules.

1 OSI Model Review (Telecom Focus)

The OSI 7-layer model provides the reference framework. In mobile/telecom networks, the most relevant layers are:

Layer	Name	Telecom Relevance
1	Physical	RF transmission, modulation, antenna design, channel coding
2	Data Link	MAC scheduling, ARQ/HARQ, radio resource management
3	Network	IP routing, GTP tunneling, mobility management
4	Transport	TCP/UDP for end-to-end delivery, SCTP in signaling
7	Application	IMS/SIP for voice, HTTP/2 in 5G SBA interfaces

Key points to remember:

- Layers 1–2 are where most wireless-specific innovation happens (PHY/MAC).
- The “user plane” carries data (Layer 3+), while the “control plane” manages signaling.
- In 5G, the protocol stack is split: CU (PDCP and above) vs. DU (RLC/MAC) vs. RU (PHY).
- GTP-U tunnels encapsulate user IP packets between base station and core.

2 Basic RF Concepts

Frequency & Bandwidth

- **Frequency (f):** Number of oscillations per second (Hz). Mobile bands range from 700 MHz to 100 GHz.
- **Wavelength:** $\lambda = c/f$. Higher frequency $\rightarrow$ shorter wavelength $\rightarrow$ smaller antennas but higher path loss.
- **Bandwidth (BW):** Range of frequencies occupied by a signal. More bandwidth $\rightarrow$ higher data rate (Shannon).

Modulation

- **Purpose:** Map digital bits onto analog carrier signals.
- **Common schemes:** BPSK, QPSK, 16-QAM, 64-QAM, 256-QAM.
- Higher-order modulation = more bits/symbol but requires better SNR.

Signal-to-Noise Ratio (SNR)

- $SNR = P_{\text{signal}}/P_{\text{noise}}$ (linear) or $10 \log_{10}(P_{\text{signal}}/P_{\text{noise}})$ in dB.
- Shannon capacity: $C = BW \cdot \log_2(1 + SNR)$ [bits/s].
- Determines which modulation and coding scheme (MCS) can be used.

Quick Reference

Modulation	Bits/symbol	Typical use
QPSK	2	Robust, used at the cell edge
64-QAM	6	High throughput, needs high SNR
256-QAM	8	Peak rates, very close to the base station

3 Digital Communications Basics

Bit Error Rate (BER)

- Probability that a received bit differs from the transmitted bit.
- Depends on SNR, modulation order, and coding.
- Typical targets: 10^{-3} (voice), 10^{-6} (data before ARQ).

Channel Coding

- **Purpose:** Add redundancy to detect/correct errors.
- **Turbo codes:** Used in 3G/4G for data channels.
- **LDPC codes:** Used in 5G NR for data (eMBB).
- **Polar codes:** Used in 5G NR for control channels.
- **Code rate (R):** Ratio of information bits to total bits. $R = k/n$ (e.g., $R = 1/2$ means 50% redundancy).

OFDM Concept

- **Problem:** Wideband channels suffer frequency-selective fading.
- **Answer** Divide wideband channel into many narrow subcarriers, each experiencing flat fading.
- **Key parameters:**
 - Subcarrier spacing (Δf): 15 kHz (LTE), 15/30/60/120/240 kHz (5G NR).
 - FFT size: determines number of subcarriers.
 - Cyclic prefix (CP): guard interval to combat inter-symbol interference (ISI).
- **Advantages:** Simple equalization (1-tap per subcarrier), flexible resource allocation.
- **OFDMA:** Multiple users share subcarriers → basis of LTE & 5G downlink.

4 IP Networking Fundamentals

Addressing

- **IPv4:** 32-bit addresses (e.g., 192.168.1.1), NAT common in mobile networks.
- **IPv6:** 128-bit addresses, increasingly used in 5G (avoids NAT).
- **Subnetting:** CIDR notation (e.g., 10.0.0.0/24 = 256 addresses).

Routing

- Mobile core networks use IP routing between network functions.
- Static routes** for simple topologies; **BGP/OSPF** in carrier networks.
- GTP tunneling**: Encapsulates subscriber IP packets inside GTP/UDP/IP between gNB and UPF.

Transport Protocols

Protocol	Properties	Use in Telecom
TCP	Reliable, ordered, congestion-controlled	HTTP/2 (5G SBA), web traffic
UDP	Unreliable, low-latency	VoIP/RTP, GTP-U user plane
SCTP	Multi-homing, message-oriented	S1AP/NGAP signaling (4G/5G)

DNS & NAT

- UEs obtain IP via PDN/PDU session establishment (not DHCP directly).
- DNS resolution for service discovery in 5G SBA (NRF).

5 Basic Probability

Why Probability in Telecom?

- Random access (RACH): collision probability with multiple users.
- Scheduling: modeling traffic arrivals (Poisson process).
- Fading: Rayleigh/Rician distributions model multipath.
- BER analysis: Q-function, error probability calculations.

Key Concepts

- Random variable (X)**: Outcome of a random experiment.
- PDF / PMF**: Probability density/mass function.
- Expected value**: $E[X] = \sum_i x_i P(x_i)$ — average outcome.
- Variance**: $\text{Var}(X) = E[(X - \mu)^2]$ — spread around the mean.

Distributions You'll Encounter

Distribution	Use Case
Uniform	Random backoff timers
Exponential	Inter-arrival times (memoryless)
Poisson	Number of arrivals in a time window
Rayleigh	Envelope of multipath fading (NLOS)
Gaussian (Normal)	Noise modeling (AWGN)

Useful Formula

- Collision probability in slotted ALOHA with n users, 1 slot: $P(\text{collision}) = 1 - (1 - 1/K)^n \approx 1 - e^{-n/K}$ for large K .

6 Decibel Math

Why Decibels?

Wireless signals span enormous dynamic ranges (pW to W). Decibels compress this into manageable numbers and turn multiplications into additions.

Definitions

Unit	Formula	Reference
dB	$10 \log_{10}(P_1/P_2)$	Ratio (dimensionless)
dBm	$10 \log_{10}(P/1 \text{ mW})$	Absolute power ref. to 1 mW
dBW	$10 \log_{10}(P/1 \text{ W})$	Absolute power ref. to 1 W

Common Conversions

Linear	dB
$\times 2$	+3 dB
$\times 10$	+10 dB
$\times 100$	+20 dB
$\times 1000$	+30 dB
$\times 0.5$	−3 dB
$\times 0.1$	−10 dB

Quick Reference Table

Power	dBm
1 mW	0 dBm
10 mW	10 dBm
100 mW	20 dBm
1 W	30 dBm
20 W (macro BS)	43 dBm

Link Budget (Preview)

$$P_{\text{Rx}} [\text{dBm}] = P_{\text{Tx}} [\text{dBm}] + G_{\text{Tx}} [\text{dBi}] - L_{\text{path}} [\text{dB}] + G_{\text{Rx}} [\text{dBi}]$$

(received power = transmit power + transmit-antenna gain − path loss + receive-antenna gain)

Practice Problems

1. A transmitter outputs 2 W. Express in dBm. $\rightarrow 10 \log_{10}(2000) = 33 \text{ dBm}$.
2. Path loss is 130 dB, Tx = 43 dBm, antenna gains total 18 dBi. Rx power? $\rightarrow 43 + 18 - 130 = -69 \text{ dBm}$.
3. SNR = 20 dB. What is the linear ratio? $\rightarrow 10^{20/10} = 100$.

Summary Checklist

Before starting Module 2, verify you can:

- ☐ Identify which OSI layers handle PHY, MAC, and IP functions
- ☐ Convert between linear power and dBm
- ☐ Explain why OFDM combats frequency-selective fading
- ☐ Calculate Shannon capacity given BW and SNR
- ☐ Describe the difference between TCP and UDP and where each is used
- ☐ Compute simple collision probabilities for random access

If any topic feels unfamiliar, review the referenced sections or consult the recommended textbooks in the syllabus.

Last updated: August 2026